

What the new exploratory analysis shows

Household electricity demand, before the Bayesian model

Project 17

October 3, 2026

Abstract

The first pass at the data checked the response scale and the residual autocorrelation. This note is the deeper pass: where the day-to-day variance sits, why August looks so low, which circuits carry the winter peak, and why the weekend effect is not one number. The fits below are ordinary least squares. Intervals are descriptive Newey–West intervals, not the posteriors in the main report. The same material is written into the technical report; this PDF is the reading version.

1 What was added

The new script, `src/01d_indepth_eda.py`, stays on the daily series already built from the minute data: 1,417 days with at least 95% of their minutes observed, from 17 December 2006 to 25 November 2010. It adds five things the earlier checks did not do.

1. A sequential variance split on the kWh scale: trend, then each annual harmonic, then day of week, with month indicators as a ceiling.
2. The shape of the year, year by year, against one harmonic and against two, together with the within-month spread.
3. A split of each day into the three sub-meters and the unmetered remainder, by month and for the August 2008 vacancy.
4. The weekly pattern after month effects, including a separate weekend-minus-weekday contrast in each season, and the average power through the hours of a winter day and a summer day.
5. A trend slope under several cuts of the sample, and a check that French public holidays are not a missing mean term.

The dependence checks in this pass are on the kWh scale, which is the scale the held-out forecasts later prefer.

2 Six findings

2.1 The calendar explains under half of a day, and almost all of that is one wave

Specification	R^2	Increment	Residual SD
Intercept only	0.000	0.000	10.00
Linear trend	0.011	0.011	9.94
Add first harmonic	0.375	0.364	7.90
Add second harmonic	0.400	0.024	7.75
Add day of week	0.452	0.052	7.40
Add third harmonic	0.467	0.015	7.30
Add fourth harmonic	0.478	0.011	7.22
Trend + month + day of week	0.470	—	7.28

Each row adds the named terms to the rows above it. The last row replaces the harmonics with a separate level for each month; it is a ceiling, not a further step.

A linear trend adds $R^2 = 0.011$. The first harmonic adds 0.364, which is four fifths of the $R^2 = 0.452$ reached by trend, two harmonics, and day of week together. The second harmonic adds 0.024. Day of week adds 0.052. The residual standard deviation is then 7.40 kWh, against 10.00 kWh with no structure at all. A bit over half of the variation from day to day is still left over. That is why the predictive intervals in the main analysis are wide: the calendar is real, and a typical day still sits several kWh off it.

The third and fourth harmonics add 0.015 and 0.011. Letting every month have its own level, on top of the trend and day of week, reaches $R^2 = 0.470$, only 0.018 above two harmonics. Two harmonics already carry the season. What they miss is a small lack of fit.

A single weekend indicator, after month effects, adds 0.046. Giving the other weekdays their own levels adds only 0.007 more. The weekend is the weekly variance. Monday and Thursday are still low, by about 2–3 kWh, but that pattern is a small slice of R^2 .

2.2 August's mean of 13.7 kWh is a mixture

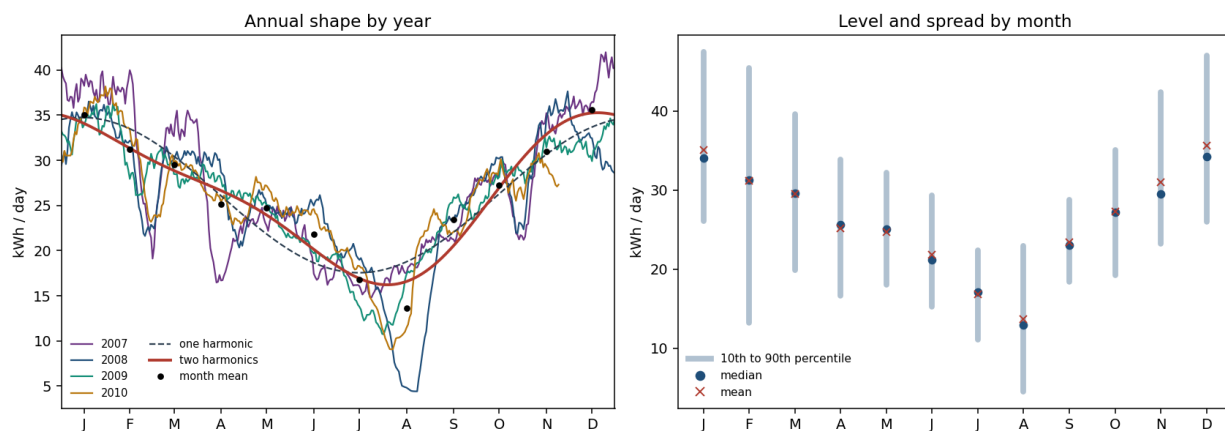


Figure 1: Left: a 15-day moving average in each year, the month means, and the seasonal curve with one harmonic and with two. The 2008 collapse in August is the vacant stretch. Right: within each month, the 10th to the 90th percentile, the median, and the mean.

With one harmonic the fitted curve peaks on 14 January at 34.8 kWh and troughs on 16 July at 17.6 kWh: a range of 17.2 kWh, amplitude 8.6 kWh. Two harmonics move the crest to 20 December and the trough to 3 August, and stretch the range to 19.1 kWh. The second harmonic's own amplitude is 2.2 kWh. The gain is the shape: a later, deeper summer floor, and a crest that sits in late December rather than a symmetric half-year away from mid-July.

The same shape repeats. Correlations of the twelve monthly means between years are 0.89 to 0.93, except 2007 against 2008 at 0.77, and that exception is August 2008.

Of 117 retained August days, 47 fall in a run of at least five days under 12 kWh. The other 70 average 18.30 kWh (median 18.18). July averages 16.81 kWh. An occupied summer is about 17–18 kWh a day. The published August mean of 13.67 kWh, and the August median of 12.93, sit that low because the vacancies are about two fifths of the August sample, enough to reach the middle. The 10th percentile is 4.54 kWh and the 90th is 22.91. The two-harmonic trough, 16.2 kWh, falls between the raw August mean and the occupied-August mean. The notch to about 5 kWh on the 2008 curve is an empty house, not a second climate.

From 6 to 30 August 2008 the house averaged 4.76 kWh a day, which is 0.20 kW running continuously. The kitchen contribution that fortnight is zero. July 2008 averaged 19.07 kWh, and August 2007, the August with no long vacancy, averaged 18.33 kWh. Those two match.

December needs a smaller version of the same caution. The fifteen days of December 2006 average 45.1 kWh; later Decembers average 34.1. The record opens on its highest fortnight. The maximum, 79.6 kWh on 23 December 2006, is in it.

Public holidays are not a missing piece of the mean. Across 43 retained holiday dates, the residual after month and day-of-week means is +0.4 kWh. Christmas is high in two years and low in 2008. The last week of December runs 2.2 kWh above the rest of December, with a standard error of 2.5.

2.3 The winter peak is heating and hot water

End use	Mean	Share	Dec–Aug	Share of gap
Unmetered	13.41	51.2%	+14.20	64.6%
Water heater and air conditioner	9.30	35.5%	+5.67	25.8%
Laundry	1.87	7.1%	+0.94	4.3%
Kitchen	1.61	6.2%	+1.17	5.3%
Total	26.19	100%	+21.97	100%

Just over half of a typical day, 13.41 kWh of 26.19, is on none of the three sub-meters. That remainder is lighting, electronics, and — in this house — heating. Sub-meter 3, which the dataset labels as the water heater and an air conditioner, is 9.30 kWh a day, 35% of the total. Laundry is 1.87 and the kitchen is 1.61.

The December-minus-August gap is 22.0 kWh a day if every August day is kept, vacancies included. Of that gap, 65% is unmetered and 26% is sub-meter 3. Kitchen and laundry together are about 10%. Drop the vacancy blocks and December still sits 17.3 kWh above August, of which 12.3 kWh is unmetered. The annual cycle is heating and hot water.

Sub-meter 3 itself falls from 11.2 kWh a day in January to 6.2 in July. An air conditioner would rise in summer. In this house the circuit falls as the weather warms, which is what an electric water heater does when the inlet water is warmer. The air-conditioner half of the label does not show up in the seasonal totals.

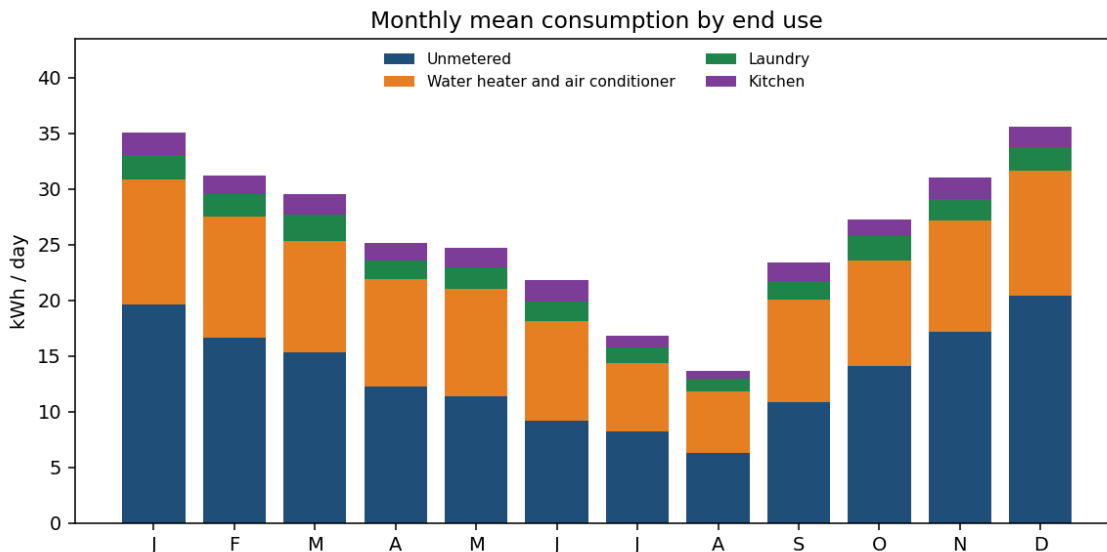


Figure 2: Monthly mean by end use. The unmetered remainder and the water-heater circuit move with the season. Kitchen and laundry stay small in every month.

2.4 The weekend effect is about 8 kWh in winter and about 2 kWh in summer

Contrast	kWh/day	95% interval
Mon	-2.19	[-2.93, -1.46]
Tue	-0.55	[-1.35, +0.25]
Wed	-0.23	[-1.07, +0.61]
Thu	-2.69	[-3.48, -1.90]
Fri	-1.12	[-1.88, -0.36]
Sat	+3.69	[+2.69, +4.70]
Sun	+3.09	[+2.03, +4.15]
Winter weekend – weekday	+7.99	[+5.24, +10.73]
Spring weekend – weekday	+5.24	[+3.18, +7.31]
Summer weekend – weekday	+2.14	[+0.79, +3.48]
Autumn weekend – weekday	+3.70	[+1.83, +5.56]

The first seven rows are deviations from the seven-day mean, after month indicators. Saturday is +3.69 kWh and Sunday +3.09. Monday is -2.19 and Thursday -2.69. Tuesday and Wednesday are compatible with the weekly mean. The pooled weekend premium is +4.75 kWh a day.

That pooled number is an average of two different seasons. In winter the weekend premium is +7.99 kWh a day, with interval [+5.24, +10.73]. In summer it is +2.14 kWh, with interval [+0.79, +3.48]. Winter's interval lies entirely above summer's. The additive day-of-week term in the Bayesian model reports the average of the two. A winter Saturday sits well above it.

The circuits say why. In winter, Saturday averages 40.7 kWh and Monday 30.5. Of the 10.1 kWh gap, 4.4 is unmetered, 2.1 is laundry, 1.9 is the kitchen, and 1.7 is the water heater. In summer the same Saturday-minus-Monday gap is 2.2 kWh, and the unmetered channel contributes 0.3 of it. Summer weekends are extra cooking and laundry. Winter weekends are those, plus the heating, because someone is home.

Winter laundry has its own routine: about 2.9 kWh on Wednesday and 3.0–3.3 at the weekend, against about 1 kWh on Monday and Thursday. That accounts for a large part of Thursday's winter

deficit relative to Wednesday. The data do not identify a calendar cause beyond that routine.

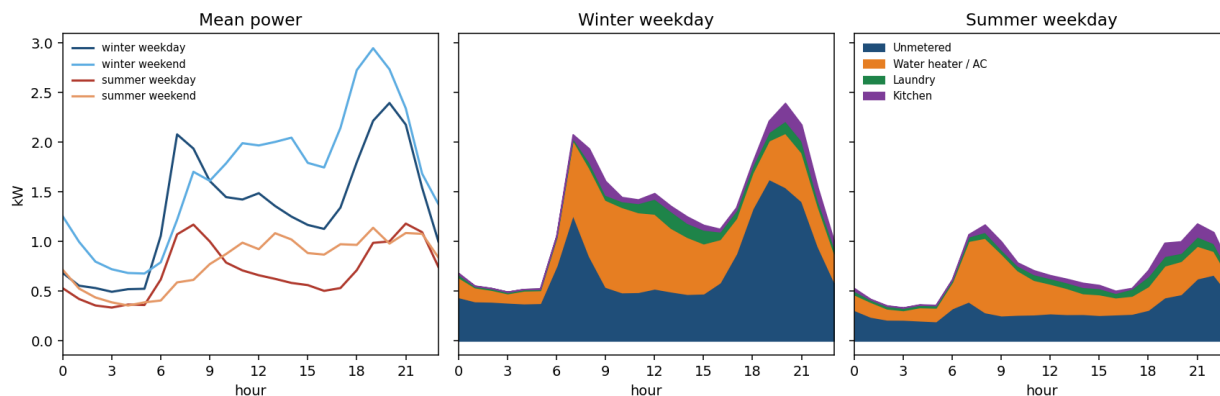


Figure 3: Left: mean power by hour in winter (December–February) and summer (June–August). Centre and right: the weekday profiles split by end use. A winter weekday integrates to 31.7 kWh and a summer weekday to 16.9 kWh.

The hour of day is the same fact, finer. A winter weekday integrates to 31.7 kWh and a winter weekend day to 39.7; a summer weekday integrates to 16.9 and a summer weekend day to 18.9. The differences, 8.0 and 2.0 kWh, match the regression. Overnight, a winter weekday sits at 0.52 kW and a summer weekday at 0.37 kW. The evening peak is 2.39 kW on a winter weekday and 1.18 kW on a summer weekday.

At 07:00 on a winter weekday the load is 2.08 kW, of which 1.27 is unmetered and 0.75 is the water heater. At 07:00 on a winter weekend it is 1.22 kW: the house wakes later. The summer spike at 07:00 is 1.07 kW, and 0.61 of that is the water heater. Heating dominates the winter peaks. The water heater dominates what is left of a summer morning.

Missing minutes are spread evenly across the clock, between 1.0% and 1.4% of rows in every hour, so this shape is not an hour at which the meter drops out. The 23 discarded days form eight short runs. The longest is 17–22 August 2010.

2.5 Any trend is a few tenths of a kWh per year

Sample	n	Slope (kWh/year)	95% interval
All retained days	1417	-0.358	[-0.976, +0.260]
Drop 2006	1402	-0.205	[-0.799, +0.389]
Drop low-use blocks	1360	-0.427	[-1.005, +0.151]
Complete years 2007-2009	1088	-0.507	[-1.382, +0.368]

The slope is the coefficient on $t/365$, with month and day-of-week indicators in the same regression, so the annual cycle is saturated rather than left to two harmonics. Every cut has a negative point estimate, between -0.21 and -0.51 kWh per year, and every interval covers zero. Dropping December 2006 moves the estimate from -0.36 to -0.21 , because that fortnight is high and it sits at the start of the series. The sign is negative either way. Four years of one house cannot separate a drift of this size from no drift.

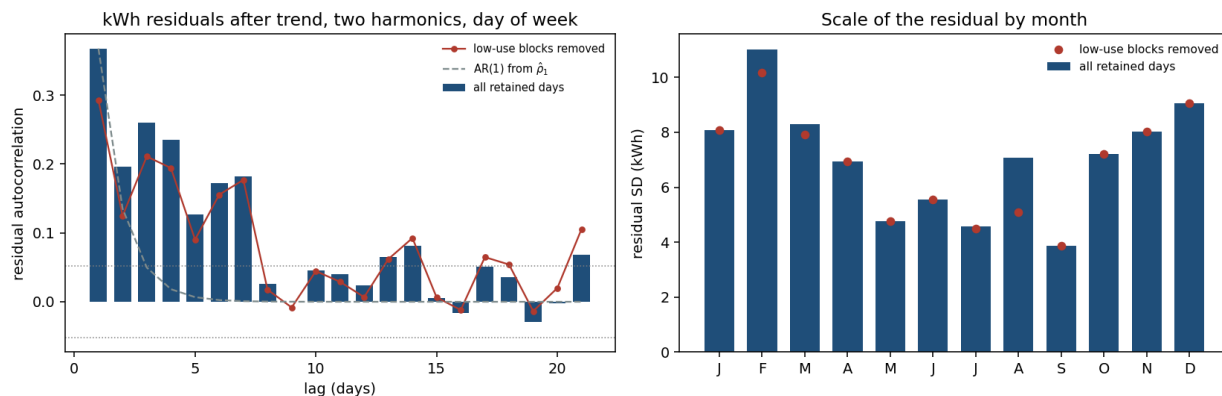


Figure 4: Residuals after a trend, two harmonics, and day of week, on the kWh scale. Left: autocorrelation, with the low-use blocks removed in red, against the geometric decay an AR(1) would produce. Right: residual standard deviation by month. August drops once the vacancies are removed. February does not.

2.6 On the kWh scale the memory is short, and both tails are heavy

The earlier autocorrelation plot was on the log scale, where lag 1 is 0.58 and the slow decay is largely the vacancy blocks. On the kWh scale, lag 1 is 0.37. An AR(1) with that coefficient makes a long-horizon band only about 7% wider than a one-day-ahead band. That is the descriptive version of the forecast result: knowing yesterday helps for a few days and then stops.

Removing the vacancy blocks lowers lag 1 to 0.29. The rest is still not a geometric decay. Lag 3 stays near 0.21. Lag 7 stays near 0.18 in the autocorrelation, but its partial autocorrelation is about 0.05, so little of it is a weekly cycle left over after the day-of-week terms.

The heavy tail is real, and it is broader than the four long absences. Residual kurtosis is about 5, against 3 for a Gaussian. The four blocks are 4% of days and 11% of the residual sum of squares. The worst 5% of days hold 40% of that sum of squares, split about evenly between winter spikes and short low days. Only 11 of those 71 extreme days sit inside the long blocks. The largest single residual is 23 December 2006, at +39.8 kWh. A t innovation has both tails to absorb.

The spread is also seasonal. After the mean structure, winter residuals have standard deviation 9.4 kWh and summer residuals 6.1; without the blocks the two numbers are 9.1 and 5.1. February stays near 10 kWh. September is 3.9 kWh. A single innovation variance averages a volatile February and a quiet September.

3 What this changes in the model, and what it does not

The deeper look supports the choices the holdout already made, and it names three approximations those choices leave in place.

The kWh scale is the right one to model on. Logging turns the vacancy floor into extreme left outliers and inflates the apparent memory. Two harmonics are enough to carry the season; they are not enough to reproduce every wiggle, but the missing piece is small in R^2 . An AR(1) is a fair summary of the short-run dependence. A heavy-tailed innovation is earned by the winter spikes as well as by August.

Three things an additive model averages over, and which the exploratory analysis can now state in kWh:

- The weekend premium is about 8 kWh a day in winter and about 2 kWh a day in summer.

- August's raw mean mixes an occupied summer near 18 kWh with an empty house near 5 kWh. A day-of-year climatology trained through 2009 folds the 2008 vacancy into mid-August, and the test year is vacant over some of the same dates.
- Winter days are more variable than summer days even after the vacancies are removed, and February in particular still looks like weather rather than occupancy. There is no temperature series in the model.

The numbers in the main report's posterior summaries remain the ones to quote for the scientific questions. This note is what those summaries are describing.